AD 502 – Machine Learning

Pre-Reading Material – Lecture 1
Introduction to Machine Learning

Estimated Reading Time: 20–25 minutes
Reference Reading: Pedro Domingos, 'A Few Useful Things to Know About Machine Learning', Communications of the ACM (2012).

Why This Topic Matters

Machine Learning (ML) has become one of the most influential technologies of the 21st century. From healthcare and finance to transportation and power systems, ML enables computers to learn from historical data and make intelligent predictions. As future engineers, understanding the basic principles of ML will help you appreciate how intelligent systems are designed and deployed.

During the first lecture, we will focus on the motivation behind Machine Learning rather than mathematical details. This pre-reading introduces the ideas that will be discussed in class and prepares you to actively participate in classroom discussions.

Learning Outcomes

After completing this pre-reading and attending the lecture, you should be able to:
- Define Machine Learning in your own words.
- Differentiate traditional programming from Machine Learning.
- Identify common applications of ML.
- Explain why data is central to learning.
- Recognize why no single algorithm is best for every problem.

Activate Your Prior Knowledge

1. How does YouTube recommend videos?

Every interaction such as watching, liking, searching, or skipping a video generates data. Think about what information YouTube might collect and how those patterns could be used to recommend videos that match your interests.

2. How does Gmail detect spam?

Email providers analyse millions of previously classified emails to learn patterns associated with spam. While reading, consider how learning from examples differs from relying only on manually written rules.

3. Can a computer improve without explicit reprogramming?

Traditional software follows fixed instructions, whereas Machine Learning systems improve by analysing new data. Reflect on how this capability makes modern applications more adaptive.

4. What data does Google Maps use?

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Traffic conditions, GPS locations, historical travel times and user reports help predict routes and arrival times. Consider why combining different sources of data leads to better predictions.

Prerequisite Concepts

You are expected to be familiar with basic programming concepts, variables, loops, functions and simple graphical representations of data. No prior knowledge of Machine Learning algorithms is required; this lecture focuses on understanding the motivation, terminology and importance of learning from data.

Core Concepts

What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data instead of relying solely on explicitly programmed rules. A trained model uses these learned patterns to make predictions or decisions for new, unseen data.

Traditional Programming vs Machine Learning

In traditional programming, programmers define every rule needed to solve a problem. In Machine Learning, the system discovers relationships from examples, making it suitable for problems where writing explicit rules is difficult.

Learning from Data

The quality, quantity and relevance of data strongly influence the performance of a Machine Learning model. High-quality data often contributes more to success than choosing a highly sophisticated algorithm.

No Universal Best Algorithm

Different problems require different learning strategies. An algorithm that performs exceptionally well for image recognition may not be suitable for predicting electricity demand or detecting financial fraud.

Generalization

A successful Machine Learning model should perform well not only on the training data but also on new unseen data. Models that simply memorize training examples suffer from overfitting and usually perform poorly in practice.

Guided Reading Questions

1. Why has Machine Learning become important?

Identify the technological and societal factors discussed by the author that have contributed to the rapid growth of Machine Learning across different industries.

2. Why is data often more valuable than algorithms?

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Notice how the article explains the relationship between data quality and model performance. Consider why improving data can sometimes produce larger gains than changing the algorithm.

3. What does 'no single best algorithm' mean?

Think about why different datasets and problems require different approaches instead of a universal solution.

4. Why should models be tested on unseen data?

Observe how testing helps measure whether a model has genuinely learned useful patterns rather than memorizing the training examples.

5. What is overfitting?

Look for the explanation of why memorizing training data leads to poor performance on new data and why avoiding overfitting is essential.

Reflection Activity

Write a 150–200 word reflection describing one Machine Learning application that you use regularly. Explain what type of data the application might collect, what prediction or decision it makes, and how its performance could improve as more data becomes available.

Self-Assessment

Answer the following before class:
1. Machine Learning primarily learns from ______.
2. Give three real-world applications of Machine Learning.
3. Why is data quality important?
4. What is meant by generalization?
5. Why is there no single best Machine Learning algorithm?

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